A Modular Edge-IoT Architecture for Smart Poultry Farm Monitoring

A design-oriented study of low-cost sensing, reliable alerts, and optional computer vision

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Abstract

Small poultry farms can benefit from continuous monitoring, but practical deployments must balance sensing capability, reliability, cost, and ease of maintenance. This paper presents a modular architecture for a low-cost smart poultry monitoring system built around environmental and resource-level sensors, a microcontroller gateway, a server-side data service, and a farmer-facing dashboard. The design emphasizes calibrated measurements, local safety rules, fault visibility, and graceful operation during network interruptions. An optional computer-vision module is discussed as a later extension, with attention to dataset bias, uncertainty, and human oversight. Rather than claiming experimental performance, the paper defines a staged evaluation plan covering measurement error, communication reliability, alert quality, and model metrics. The central argument is that dependable sensing and actionable alerts should precede advanced AI in small-farm automation.

Keywords: Internet of Things; edge computing; poultry monitoring; sensor calibration; computer vision; farm automation

1. Introduction

Small-scale poultry farms increasingly face pressure to improve animal welfare, reduce avoidable losses, and use labor more efficiently. Environmental monitoring and routine inspection are often performed manually. This can make it difficult to detect gradual changes in temperature, humidity, feed availability, or animal activity before they become operational problems. Low-cost sensors and edge computing offer a practical route toward more continuous monitoring.

The term *smart poultry monitoring* covers a broad range of technologies, from simple temperature alarms to camera-based behavior analysis. Yet a system that is technically sophisticated is not automatically useful on a farm. Reliability, maintainability, connectivity, power consumption, and the clarity of alerts are at least as important as model accuracy. This paper proposes a modular architecture that places dependable sensing and transparent alert logic before advanced artificial intelligence.

2. Design Objectives and Scope

The proposed system is intended for a small poultry house with one local gateway and several sensing or actuation points. It is a design proposal rather than a report of a deployed experiment. Its primary objective is to demonstrate how sensing, communication, data storage, and decision support can be separated into components that can be tested independently.

- **Observability:** collect a small set of interpretable measurements, including air temperature, relative humidity, feed level, and water level.
- **Actionability:** translate measurements into understandable notifications and, where safe, bounded control actions.
- **Resilience:** preserve local operation when the internet connection or remote server is unavailable.
- **Extensibility:** allow camera-based disease or behavior analysis to be added without redesigning the basic monitoring layer.

3. Proposed System Architecture

3.1 Sensing layer

A practical prototype may use a digital temperature-and-humidity sensor in the animal area, load cells beneath feed or water containers, and ultrasonic distance sensors where container geometry makes distance-to-level estimation suitable. Sensor placement matters: a temperature sensor mounted near a heater, a fan outlet, or direct sunlight may not represent the conditions experienced by the flock. Measurements should therefore be collected at representative locations and protected from pecking, dust, and moisture.

Feed and water estimates require calibration. For a load cell, the raw analog-to-digital converter reading should be mapped to mass using known reference weights. For an ultrasonic sensor, the measured distance must be converted using the container's geometry; irregular container shapes may require a lookup table or piecewise calibration. The system should report sensor faults and implausible readings rather than silently presenting them as valid values.

3.2 Edge gateway and communication

A microcontroller such as an ESP32 can serve as the local gateway for a small prototype. It reads sensors, applies basic filtering, checks safety limits, and forwards time-stamped records to a server using an authenticated HTTP endpoint or a lightweight messaging protocol. A gateway should retain the most recent readings locally and queue a limited number of records during temporary network outages. Reconnection logic should use bounded retries and avoid flooding the server.

A star topology is appropriate for a first prototype: sensor nodes communicate with one gateway, and the gateway communicates with the server. The choice between wired links, Wi-Fi, and low-power radio depends on distance, interference, power availability, and the physical layout. Communication should be tested in the actual coop, because walls, metal structures, and electrical equipment can reduce radio reliability.

3.3 Server and application layer

The server receives measurements, validates their format, stores them with device identifiers and timestamps, and exposes them through an API. A relational database is sufficient for an initial deployment if the schema is designed around devices, locations, measurements, alerts, and user actions. A web dashboard can show current conditions, recent trends, device status, and alert history. The interface should distinguish a missing reading from a normal reading and should show when each value was last updated.

4. Monitoring and Control Logic

A robust first release should favor simple, inspectable rules over a complex predictive model. For example, a temperature alert can be triggered when a measured value remains above a configurable threshold for a defined duration. Hysteresis can prevent repeated on-off switching around a boundary. A separate hard safety limit should override ordinary user settings when the hardware supports such protection.

Actuators such as fans, lamps, or feeders introduce physical risk. Their control should include maximum run times, manual override, safe startup states, and independent electrical protection. Software must not be treated as a substitute for correctly rated relays, fuses, enclosures, grounding, and qualified electrical installation. A remote command should be acknowledged, logged, and rejected if it falls outside the allowed operating range.

5. Optional AI Extension

Computer vision can be added to analyze selected images from a coop camera. A detector may identify visible animals or unusual visual conditions, while a classifier may assess a narrowly defined condition if an appropriately labeled dataset exists. However, a model trained on one breed, lighting setup, camera angle, or farm may perform poorly elsewhere. Dataset documentation should record farm conditions, image capture procedures, class definitions, and known gaps.

AI outputs should be treated as decision-support signals rather than definitive diagnoses. Low-confidence predictions should be marked as uncertain, and high-impact decisions should require human inspection. Evaluation should report per-class precision, recall, F1-score, and a confusion matrix, not only overall accuracy. If disease detection is a goal, veterinary expertise is needed to define labels and verify the practical meaning of predictions.

6. Evaluation Plan

A staged evaluation can establish whether the system is useful before introducing machine learning. The first stage checks sensor calibration against reference instruments. The second measures data delivery, missing records, reconnection behavior, and battery or power consumption. The third evaluates alert behavior using controlled test conditions. Only after these stages should an AI module be evaluated.

Evaluation dimension	Suggested measure	Purpose
Measurement quality	Mean absolute error; drift over time	Check whether readings are dependable
Communication	Delivery rate; end-to-end latency; outage recovery	Assess reliability under farm conditions
Alert quality	False-alert rate; missed-event rate; time to alert	Measure operational usefulness
AI module	Precision, recall, F1-score by class	Expose class-specific strengths and weaknesses

7. Discussion

The main engineering trade-off is between functionality and operational complexity. Adding more sensors can improve visibility, but each sensor adds calibration, maintenance, and failure modes. A modular design allows a farm to begin with a few high-value measurements and expand only when the added information changes a real decision. Similarly, cloud dashboards improve remote access but should not become a single point of failure for essential local safety functions.

Data quality and human workflow are central concerns. Alerts that are too frequent may be ignored; alerts without a clear recommended action may create additional work. The system should therefore be co-designed with farm operators, use familiar units and language, and provide a concise explanation for each alert. Data retention and access controls should also be defined, especially if camera images or identifiable worker activity are collected.

8. Limitations

This manuscript presents a conceptual architecture and evaluation plan. It does not claim measured improvements in productivity, animal health, or model performance. Actual sensor selection and thresholds depend on local husbandry practice, coop design, climate, and applicable animal-welfare guidance. Field validation across multiple farms and seasons would be required before making general performance claims.

9. Conclusion

A practical smart poultry monitoring system can be developed incrementally by separating sensing, edge processing, communication, storage, visualization, and optional AI. The strongest starting point is not the most complex model, but a reliable measurement and alerting loop that farmers can understand and maintain. Future work should implement the proposed prototype, conduct calibration and reliability tests, and assess whether AI-based visual signals provide measurable value beyond well-designed sensor rules.

References

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Editorial note: This is an original, design-oriented educational manuscript. It is not a report of a completed field trial and has not undergone peer review. References are provided for background and technical context.